Additional file 2: Anchor theory

1. Object of study and interpretation

Let D and E be finite persistence diagrams with finite coordinates, including the diagonal with unlimited multiplicity in their bottleneck comparison. Write $E + t$ for translation of both coordinates by t and define

$$f(t) = d_B(D, E + t).$$

For the present superlevel contact construction, implemented as a sublevel filtration of $-L$, subtracting a map mean from L adds that mean to its persistence coordinates. Consequently, with $\delta = \text{mean}(L_{alt}) - \text{mean}(L_{ref})$, the native and mean-centered distances are $f(0)$ and $f(\delta)$. Common translation of both diagrams cancels. Every individual interval lifetime is unchanged by this operation.

This identity does not say that a centered score is a function of the native scalar distance and mean change alone. For $D = \{(0, 4)\}$, $E_+ = \{(1, 5)\}$ and $E_- = \{(-1, 3)\}$, both native distances equal 1. Under the same $\delta = 1/4$ their distances become $5/4$ and $3/4$. The direction is information about diagram location lost in the native scalar summary. Appropriate interpretation: **anchor-location sensitivity of a diagram comparison**. Inappropriate interpretation: proof that mean intensity alone explains a biological effect, or that centered diagrams lack structural information.

2. Endpoint rigidity and an error-aware path tube

Translation and the triangle inequality give $|f(t) - f(s)| \le |t - s|$. This elementary inequality yields a stronger diagnostic than checking selected intermediate shifts.

Let $\delta \neq 0$, $T = |\delta|$, $h = \text{sign}(\delta)$, and $g(u) = f(hu)$, $0 \le u \le T$. Put $A = g(0)$, $C = g(T)$. For every u,

$$\max\{A - u, C - (T - u)\} \le g(u) \le \min\{A + u, C + (T - u)\}. \qquad (1)$$

For either $r \in \{-1, +1\}$, define the saturation deficit

$$\kappa_r = T - r(C - A) \ge 0.$$

Then, throughout the entire segment,

$$u - \kappa_r \le r[g(u) - A] \le u. \qquad (2)$$

Proof. Apply the 1-Lipschitz inequality from the starting endpoint and from the ending endpoint, obtaining (1); substitute $C - A = r(T - \kappa_r)$. Both signs give (2). No assumption about a fixed optimal matching is needed.

If $|C - A| = T$, taking $r = \text{sign}(C - A)$ gives $\kappa_r = 0$ and therefore $g(u) = A + ru$ everywhere on the segment. This is an exact theorem for any shift magnitude, not just an infinitesimal result. It does **not**

imply that one matching remains optimal, or that this affine formula extends outside the segment.

Suppose only estimates $\widehat{A}$, $\widehat{C}$ are available, with proven bounds $|\widehat{A} - A| \le e_0$ and $|\widehat{C} - C| \le e_T$. Then

$$\kappa_r \le K_r := T - r(\widehat{C} - \widehat{A}) + e_0 + e_T. \qquad (3)$$

For valid error bounds K_r is nonnegative. Combining (2) and (3), the whole path lies within $e_0 + K_r$ of the line $\widehat{A} + ru$. A sharper signed statement is $r[g(u) - A] \in [u - K_r, u]$. For nonsaturated paths, the following envelopes need no affine approximation:

$$L(u) = \max\{0, \widehat{A} - e_0 - u, \widehat{C} - e_T - (T - u)\},$$
$$U(u) = \min\{\widehat{A} + e_0 + u, \widehat{C} + e_T + (T - u)\}.$$

They satisfy $L(u) \le g(u) \le U(u)$. At $\delta = 0$ there is no path to audit.

There are two distinct computational claims. A residual at machine precision is **observed numerical saturation**. A certified K requires endpoint error bounds covering every relevant operation: PH input conversion, coordinate recovery and translation, mean estimation when an exact-real mean is the target, distance evaluation and its returned rounding. The existing float32 filtration audit alone does not verify all binary64 arithmetic. One can conservatively include such errors, but must state the budget and its provenance. A solver's exact algorithm mode is not, by itself, a formal interval-arithmetic proof for a real-valued source map. API quantization remains part of the delivered data and cannot be reversed by this audit.

If δ is uncertain by e_δ, either define the audit target as the explicitly stored shift, or add e_δ to the endpoint/path error when comparing to the intended shift; 1-Lipschitz continuity justifies that addition.

3. Why local slopes have only three possible values

For $x = (b_x, d_x) \in D$ and $y = (b_y, d_y) \in E$, set

$$c_{xy} = ((b_x - b_y) + (d_x - d_y))/2,$$
$$q_{xy} = |(b_x - b_y) - (d_x - d_y)|/2.$$

Their matching cost is exactly

$$\varphi_{xy}(t) = \max\{|b_x - b_y - t|, |d_x - d_y - t|\} = |t - c_{xy}| + q_{xy}.$$

Matching a point to the diagonal costs half its lifetime, independently of t. For any fixed partial matching M, the maximum cost therefore has the form $\max\{a_M - t, b_M + t, c_M\}$, allowing absent real-real terms and taking $c_M \ge 0$. There are finitely many partial matchings. Thus f is a finite lower envelope of these functions: continuous, piecewise affine, 1-Lipschitz, with affine slopes in $\{-1, 0, +1\}$. It need not be convex: for $D = E = \{(0, 2)\}$, $f(t) = \min\{|t|, 1\}$.

The familiar parameterized-matching structure is a useful derivation for this audit, not a claim of new mathematical machinery.

A native-diagram-only local certificate

The following provides a prospective slope from native diagrams, without consulting a centered endpoint or selected-allele labels. If either finite diagram is empty, f is constant under translation, with slope zero for every shift. Otherwise, assume $b = f(0) > 0$. Form the standard finite augmented bipartite graph: real D points and E-diagonal copies on the left; real E points and D-diagonal copies on the right. Include every real-real edge, each real point's own diagonal edge, and all diagonal-copy to diagonal-copy edges of cost zero. A perfect matching encodes a persistence matching.

Call an edge low, critical or high according as its native cost is $< b$, $= b$ or $> b$. For a direction $h \in \{-1, +1\}$, a critical real-real edge has one-sided slope $a_e = -1$ if $hc_{xy} > 0$, and $+1$ otherwise. Critical diagonal edges have slope zero. Define

$$\gamma = \min\{|w_e(0) - b| : w_e(0) \neq b\},$$
$$R = \min\{\gamma/2,\ |c_{xy}| \text{ for critical real-real edges with } hc_{xy} > 0\}.$$

An empty second minimum is $+\infty$. Zero-cost dummy edges ensure the first minimum exists when $b > 0$. For $0 < u < R$, low edges stay below each candidate level $b + ru$, high edges stay above, and critical edges retain their one-sided affine formulas.

Let G_r include all low edges and the critical edges with $a_e \leq r$. The smallest $r \in \{-1, 0, +1\}$ for which G_r has a perfect matching satisfies

$$f(hu) = b + ru \quad \text{for all } 0 \leq u < R. \qquad (4)$$

Proof. The stated inequalities fix which edges can be used at the threshold $b + ru$. A native-optimal matching must contain at least one critical edge, since otherwise its native maximum would be $< b$. Therefore the minimum, over optimal matchings, of their largest critical slope is the exact active slope; nonoptimal matchings cannot enter below it before the gap closes. This is equivalently the three threshold-matching tests above.

This radius is conservative and may be much smaller than an empirical mean shift. Predicting the native one-sided slope does not justify extending it beyond R. Cases $b = 0$ should use a separate degeneracy treatment, rather than apply this statement unchanged. The certificate can be implemented with exact dyadic arithmetic for stored binary coordinates, but dense graph construction is optional and potentially expensive.

If all coordinates in D and E lie on one affine lattice $a + g\mathbb{Z}$, then real-real costs are multiples of g and diagonal costs are multiples of $g/2$. Hence $\gamma \geq g/2$ and every positive $|c_{xy}| \geq g/2$, so $R \geq g/4$. This is a conservative lattice-derived local radius, not an assertion that an observed shift lies within it. Source quantization alone is insufficient to choose g after diagram-coordinate recovery and translation; the actual stored coordinates must be checked. Small shifts on a nonflat affine cell are necessarily saturated. The empirical frequency of saturation should therefore be compared with shift magnitudes, candidate-level gaps and flat/turning synthetic controls, rather than treated as a novel phenomenon.

4. Witnesses that can avoid another complete matching computation

A diagonal-augmented Hausdorff lower bound is

$$H(t) = \max\{ \max_x \min[\mathrm{dist}_\infty(x, E+t), \mathrm{life}(x)/2], \\ \max_y \min[\mathrm{dist}_\infty(y+t, D), \mathrm{life}(y)/2]\} \le f(t).$$

Use distance $+\infty$ to an empty finite set and value zero for a maximum over an empty point set. The inequality holds because each point in any matching either pays a real-point cost or its diagonal cost. Nearest-neighbor queries can evaluate H efficiently.

For an increasing path, $H(\delta) \le C \le A + T$. If $H(\delta) = A + T$, saturation is proved without solving the shifted matching problem. With an error-controlled lower estimate H_{low} and upper estimate A_{up}, the increasing deficit is bounded by $\kappa_+ \le A_{up} + T - H_{low}$. This is a witness-based audit using native diagrams and the chosen shift. It is independent of centered matching computation, but not independent of the chosen anchor or diagram geometry.

For a decreasing path the universal lower bound is $C \ge A - T$. A valid explicit shifted matching with maximum cost $\le A - T$ proves equality. More generally a matching upper bound U gives $\kappa_- = T + C - A \le T + U - A_{low}$. Hausdorff alone generally cannot establish this upper bound.

Hausdorff is not an exact replacement for bottleneck matching: D containing two copies of $(0, 4)$ and E containing one has $H(0) = 0$ but $f(0) = 2$. Multiplicity and Hall-type matching constraints matter.

5. Ranking, tolerance and partial orders

For allele a let its exactly saturated path be $B_a(\lambda) = A_a + \lambda k_a$, $0 \le \lambda \le 1$, where $k_a = r_a|\delta_a|$. In a numerically certified version, enclose this line by the endpoint-derived tube. Rank crossings of exact lines occur only where

$$(A_a - A_b) + \lambda(k_a - k_b) = 0.$$

For a top-set rule that includes scores within τ of the maximum, also include the roots where this expression equals $\pm\tau$. Retain only roots within $[0, 1]$, and handle parallel or identical lines separately. The pairwise roots partition the path into cells with a constant tolerance-defined top set. The boundaries require the inclusive convention used by the frozen analysis.

An exact native tie can break for every $\lambda > 0$ if $\tau = 0$ and slopes differ. With positive tolerance τ, that pair separates only after $\lambda|k_a - k_b| > \tau$. Thus top-credit changes can be large even when distance changes are arbitrarily small; the discontinuity belongs to the rank/tie functional, not to persistence distance. At fixed positive τ the discontinuities move to the tolerance boundary.

For arbitrary certified score intervals $[L_a, U_a]$, a exceeds b by more than τ whenever $L_a > U_b + \tau$. The set $\{a : U_a \ge \max_{b \ne a} L_b - \tau\}$ conservatively contains every possible tolerance-defined top allele. The set $\{a : L_a \ge \max_{b \ne a} U_b - \tau\}$ is guaranteed to belong to the top set for every admissible score vector. These possible and guaranteed sets are exact for independent Cartesian score intervals; with dependent errors they remain conservative. They define a robust partial ranking rather than forcing unsupported total

ranks. For a whole path, use the endpoint envelopes at $u = \lambda|\delta_a|$, or take the appropriate infima and suprema over the λ interval to certify uniform membership or pairwise ordering.

The standard anchor-blind margin guarantee is $A_i - A_j > |\delta_i| + |\delta_j| + \tau$, with numerical errors added when estimated quantities are used. A slope-aware path certificate can be tighter: minimize $(A_i - A_j) + \lambda(k_i - k_j)$ over the desired λ interval, which requires only its endpoints, then subtract the two tube errors and compare to τ. These are elementary robust-ranking consequences, not new generic ranking theorems.

Do not choose λ using selected-allele labels and treat the resulting enrichment as predeclared. Replication analyses should freeze the tolerance, error budget, endpoint-saturation statistic, ranking audit and any predictive baseline before map outcomes are inspected.

6. Rank-filtration perturbations

Assign edge ij the fraction of eligible edges with strictly greater L value, grouping ties so equal weights enter together. The resulting rank flag filtration is unchanged under every strictly increasing transformation of L (Giusti et al., 2015). It removes magnitude information more broadly than global centring. Near ties, small entrywise changes can alter ranks, so the weighted-filtration stability bound does not transfer automatically. If edge perturbations are bounded by δ, only comparisons whose original gap is at most 2δ can change. For a fixed set of m eligible edges, the rank-filtration perturbation is at most $\max_e \#\{f \neq e : |w_f - w_e| \leq 2\delta\}/m$. At exact ties no universal continuity bound tending to zero with δ exists: splitting an all-tied four-bin graph into a zero-birth perimeter and two arbitrarily later diagonals creates a rank interval of length 2/3.

An uncentred weighted filtration, a centred weighted filtration and a rank filtration answer different questions. Their comparison is more informative than labelling topology uniformly intensity-sensitive. Degree-one interval lengths also omit genomic feature identity: relabelling bins leaves the abstract diagram unchanged. Coordinate-sensitive comparisons remain necessary when loop location is the target.

7. Verification and prior-art scope

- Sheehy, Kisielius and Cavanna, *Computing the Shift-Invariant Bottleneck Distance for Persistence Diagrams*, CCCG 2018, already define optimization over diagonal shifts and study its computation. Their parameterized matching problem is directly relevant. The shift quotient and the idea of following shifted costs must not be presented as new: https://donsheehy.net/research/cavanna18computing.pdf
- Cao, Monod, Vlontzos, Schmidtke and Kainz, *Topological information retrieval with dilation-invariant bottleneck comparative measures*, published 4 July 2023, develop dilation-invariant comparisons and explicitly relate them to shift-invariant bottleneck work. This is close normalization/comparison prior art, although its transformation and retrieval task differ: https://academic.oup.com/imaiai/article/12/3/1964/7217093
- May, Krishnamoorthy and Gambill, *A Normalized Bottleneck Distance on Persistence Diagrams and Homology Preservation under Dimension Reduction*, submitted 11 June 2023, revised 29 August 2024 and published in La Matematica, develops scale normalization and stability in a dimension-reduction setting. It further excludes broad novelty claims about normalized persistence comparison: https://arxiv.org/abs/2306.06727 ; https://doi.org/10.1007/s44007-024-00130-0

The endpoint rigidity lemma follows from elementary Lipschitz geometry. The finite-cell slope statement follows from finite bottleneck matching. The interval ranking statements are standard margin arguments. No priority claim is made for these individual results. The review covered these closest identified sources, not every work on parametric matching or robust ranking.

8. Exact finite checks and their limits

scripts/astra_verify_anchor_theory.py independently enumerates all partial matchings on 250 small rational diagram pairs. It passed 498 directional local-slope cases (1,494 exact evaluations) and 4,500 endpoint-cone/deficit-tube checks, together with saturating, V-turn, diagonal-cap, opposite-slope and multiplicity controls. Output: outputs/astra_anchor_theory_verification.json.

All verification arithmetic uses Python Fraction. These finite checks support implementation and catch sign/degeneracy errors; the written proofs carry the universal statements. They are not a proof-assistant formalization and do not certify the full genomic diagram computations.